

Srikanth Kandula

✉ kandula040598@gmail.com | ☎ 219-386-0528 | [🌐LinkedIn](#)

👤 Sr. Data Engineer | Python Developer | Cloud & Big Data Expert | AI/ ML Engineer

Professional Summary

Senior Data Engineer and Python Developer with 5+ years of experience building scalable data platforms, real-time pipelines, and AI/ML systems across AWS, GCP, and Snowflake. Skilled in Spark, Kafka, Airflow, and Python microservices, with a recent focus on deploying generative AI models and automating ML workflows. I design clean, reliable, and production-ready systems that turn messy data into intelligent, actionable outcomes—always with a focus on performance, automation, and real impact.

Core Competencies

- **Programming Languages:** Deeply comfortable with Python — it's my daily driver for data, APIs, and automation. Also experienced with Scala and Java (mostly for Spark), SQL for everything data-related, plus Shell scripting and the occasional dive into R or C# when needed.
- **Cloud Platforms:** Hands-on experience building on AWS (S3, Glue, Lambda, Redshift, EMR), along with projects in GCP (BigQuery, Dataflow) and Azure (Databricks, Synapse) — I'm cloud-agnostic, but I lean toward whatever gets the job done right.
- **Big Data & Streaming:** Built and scaled pipelines using Apache Spark, Kafka, and Hadoop. Worked with Snowflake and Delta Lake for modern warehousing. Comfortable with HBase and Flink in real-time or batch-heavy use cases.
- **ETL & Orchestration:** Designed workflows using Airflow (a lot), AWS Glue, and Snowpipe. I've also worked with legacy tools like Informatica and Talend, though I prefer keeping things modular and Pythonic.
- **Machine Learning & AI:** Used scikit-learn and TensorFlow for classical ML, played with PyTorch for deep learning use cases. Worked with Snowpark for ML inside Snowflake, and MLLib for Spark-based training jobs.
- **DevOps & Infra:** Built and deployed infra using Terraform, containerized apps with Docker, and used Kubernetes where orchestration made sense. Familiar with Jenkins, Git, and CI/CD best practices that work in real teams.
- **Databases:** Worked with a mix of relational (MySQL, PostgreSQL, Oracle) and NoSQL (MongoDB, Cassandra, Redis). I try to choose what fits the use case rather than forcing one tech everywhere.
- **Data Visualization:** Built dashboards in Tableau, Power BI, and Apache Superset — and for more custom needs, I've used Plotly, Seaborn, and even raw D3.

- **Tools I Rely On:** JIRA for project tracking (when used well), Postman for API testing, Splunk and OpenSearch for logs and metrics, and MLFlow for experiment tracking and model management.

Professional Experience

Python Developer / Data Engineer | Citi Group, NY (Oct 2023 – Present)

- Architected real-time Kafka + Snowflake pipelines to process high-throughput financial transactions.
- Migrated legacy batch workflows to real-time, event-driven architecture using AWS Lambda, Step Functions.
- Built and containerized microservices in Python to serve real-time ML insights.
- Created scalable CI/CD infrastructure using Terraform, Jenkins, Docker, reducing release time by 75%.
- Led the development of production-grade ML pipelines in Python and PySpark to support real-time fraud detection, credit scoring, and regulatory reporting — pipelines processed millions of records daily with strict SLAs.
- Worked closely with data scientists and quants to bring ML models from notebooks to production — wrapped them in scalable Python APIs using FastAPI, integrated with existing systems, and monitored performance over time.
- Architected and built GenAI-powered tools using LLMs (OpenAI, LLaMA) to automate financial document analysis, internal reporting, and client communication workflows — focused on making AI usable, secure, and explainable.
- Fine-tuned language models on Citi-specific data using RAG and prompt engineering techniques to ensure outputs were compliant, accurate, and aligned with enterprise risk controls.
- Designed secure, cloud-native data solutions using AWS (S3, Glue, Lake Formation), with a strong emphasis on auditability, encryption, and access control for handling highly sensitive financial data.
- Played a key role in designing the CI/CD strategy for ML and GenAI applications — built GitLab pipelines for testing, packaging, and deploying model services into staging and production environments.
- Built custom data profiling and validation layers in Python to catch upstream data issues early — helped cut downstream model failures and improved trust in ML outputs.
- Modernized legacy batch workflows into real-time data streams with Kafka and Spark Structured Streaming, improving the speed and accuracy of real-time decision systems.
- Worked across business and engineering teams to shape Citi's GenAI strategy — balanced innovation with compliance, helping set standards for how LLMs should be used internally.
- Known for being the person who could go from concept to production — whether it was debugging weird Spark joins or helping stakeholders understand what an LLM *can't* do.
- Designed AWS S3-based data lake integrated with Athena and Glue, reducing query latency by 60%.
- Implemented fraud detection and risk scoring pipelines integrating quant models and ML.

Python Developer / Data Engineer | Centene Corporation, MO (Oct 2022 – Sep 2023)

- Built and maintained complex ETL pipelines in Python and PySpark to ingest and process healthcare claims, eligibility, and provider data — ensured high data quality while navigating messy, inconsistent source systems.
- Designed and optimized workflows in Apache Airflow to manage daily batch jobs across Medicaid and Medicare datasets — added retry logic, SLA monitoring, and alerting to keep pipelines resilient and auditable.
- Integrated member and clinical data from multiple internal and third-party systems, transforming it into clean, analytics-ready formats used by actuarial and care management teams.
- Developed internal APIs in Flask and FastAPI to expose curated data sets for downstream apps, analytics, and reporting — added role-based access control and JWT auth to meet HIPAA and internal security policies.
- Worked closely with data governance and compliance teams to implement data lineage, auditing, and encryption for PHI/PII across AWS S3, RDS, and Redshift.
- Refactored legacy SQL-based transformations into scalable Python and Spark jobs — reduced runtime from hours to minutes, saving compute costs and improving report delivery times.
- Supported data science teams by engineering feature pipelines, validating training data, and deploying model scoring scripts in production — helped deploy early detection models for member readmission risk.
- Took ownership of CI/CD workflows for data pipelines using GitLab CI and Terraform — made sure all jobs were tested, versioned, and reproducible from dev to prod.
- Regularly partnered with business analysts, clinicians, and compliance stakeholders — translated requirements into clean, automated data solutions they could rely on.
- Developed PySpark/Scala ETL jobs on Azure Databricks and integrated ML workflows using Snowpark.
- Built DAGs in Airflow to manage complex dependency chains and data validation checkpoints.
- Automated schema evolution and partitioning logic for dynamic data sources in Snowflake and S3.

Data Engineer | Cigna Healthcare, CT (Jan 2021 – Apr 2022)

- Designed and built scalable, production-ready data pipelines in Python and Spark to process high-volume claims, eligibility, and provider data across multiple states and lines of business.
- Re-architected legacy ETL workflows into modular, reusable Airflow DAGs, improving job reliability and reducing failure rates — pipelines now handle 10M+ rows daily with built-in retries and monitoring.
- Worked closely with actuarial, analytics, and clinical teams to deliver clean, validated data sets used in critical programs like value-based care analysis, risk adjustment, and utilization management.
- Created internal data APIs and microservices in Flask and FastAPI to serve patient and provider intelligence to downstream teams and applications — added detailed logging and metrics for observability.
- Collaborated with DevOps and security teams to automate infrastructure provisioning with Terraform and manage CI/CD workflows via GitLab for all data pipelines.

- Helped migrate core workloads from on-prem SQL servers to a cloud-native stack (AWS + Snowflake), reducing processing time and maintenance effort while improving scalability.
- Was the go-to person when jobs failed, data looked “off,” or something felt too slow — always took time to trace the issue to its root, fix it, and write it up for the team.
- Sat in regular syncs with business stakeholders to understand real-world pain points — then built data solutions that actually solved them, not just checked a box.
- Developed secure REST APIs to facilitate HIPAA-compliant healthcare data exchange.
- Built MLFlow pipelines for deploying predictive models on Databricks.
- Built automated data validation frameworks to ensure accuracy and consistency.
- Created OpenSearch dashboards to track system logs and performance metrics.
- Built metadata-driven ETL templates to accelerate new ingestion use cases.
- Contributed to data modeling for clinical datasets and optimized indexes in Aurora and Redshift.
- Performed impact analysis for data lineage using custom Python scripts and Airflow hooks.

Python Developer / Data Engineer | Amazon, Hyderabad, India (Sep 2019 – Dec 2020)

- Designed and built scalable data pipelines using Python and Spark to process billions of events daily for product metrics, operational insights, and customer experience improvements.
- Owned end-to-end pipeline development — from raw data ingestion off Kinesis and S3, through transformation in PySpark, to Redshift/Snowflake for analytics teams across retail and logistics orgs.
- Built internal Python libraries for standardized data validation, transformation, and schema enforcement — helped reduce onboarding time for new data sources and ensured consistent quality.
- Optimized PySpark workloads by tuning partitioning, caching, and memory management — brought down job runtimes from 45 minutes to under 10 while saving compute spend by 30%.
- Automated complex data workflows using Airflow, integrating upstream systems (DynamoDB, Aurora) with downstream machine learning and reporting layers — added alerts and metrics via CloudWatch and PagerDuty.
- Collaborated with SDEs, product managers, and scientists to build real-time and batch pipelines that powered personalization engines, search ranking models, and fraud detection tools.
- Wrote and maintained microservices in FastAPI to expose curated datasets via low-latency APIs — used IAM roles and fine-grained access control to secure sensitive customer data.
- Contributed to internal platform tools for metadata tracking, lineage, and cost attribution — made it easier for teams to discover, understand, and manage their data assets.

Education

- Bachelors in Computer Science and Engineering, Tirumala Engineering college, India, 2020
- Masters in Information Technology, St. Francis College, New York, NY, 2024